Introduction to VLSI Design, 2015

Homework 2: all four analytical problems and code

1. Input rise time and output fall time

With an initially high inverter output, the output capacitance discharges when the input rises. Its instantaneous equation is

$$C_L \frac{dV_{out}}{dt} = I_p - I_n.$$

The faster ramp (a) raises the NMOS overdrive more quickly and turns the PMOS off sooner. The slower ramp (b) leaves a weaker NMOS and a longer interval during which PMOS current opposes discharge. The net discharge current $I_n - I_p$ is therefore smaller during much of the output transition, so (b) has a longer output fall time. This is a load-current explanation; the output fall time is not simply equal to the input rise time. Once the input is fully high, the NMOS and C_L set the remaining discharge rate.

2. Clock-tree critical path

The root inverter contributes τ_g to every branch. B and D have one additional τ_g inverter; C has an additional $2\tau_g$ gate. Total route lengths are 650 μm for A–B/A–D and 400 μm for A–C. With the supplied estimate,

$$\tau_w = \tfrac{1}{2}(0.05)(30 \times 10^{-18})l^2 \text{ s}, \qquad 4\tau_g = 0.8 \text{ ns}.$$

Route	Length (μm)	τ_w (ps)	Gate delay (ns)
A-B	650	0.316875	0.4
A-C	400	0.120000	0.6
A-D	650	0.316875	0.4

Every wire estimate is much smaller than 0.8 ns and is neglected under the explicit rule. Therefore

$$\boxed{\text{Critical path A–C; delay} = 3\tau_g = 0.6 \text{ ns}.}$$

This uses the problem's per-route $rcl^2/2$ approximation. A detailed branched Elmore-delay model would include shared wire capacitances, but is neither supplied nor required, and cannot change the conclusion under the given neglect rule.

3. Mobility ratio

At $V_{in} = V_{out} = V_{DD}/2$, the conducting devices are saturated in the ideal long-channel model. Their currents must match:

$$\mu_n C_{ox}(2)(V_{DD}/2 - V_{tn})^2 = \mu_p C_{ox}(4)(V_{DD}/2 - |V_{tp}|)^2.$$

Hence the general result is

$$\frac{\mu_n}{\mu_p} = 2\left(\frac{V_{DD}/2 - |V_{tp}|}{V_{DD}/2 - V_{tn}}\right)^2.$$

Under the usual symmetric-threshold assumption $V_{tn} = |V_{tp}|$, the requested answer is $\boxed{\mu_n/\mu_p = 2}$. The threshold equality is needed; midpoint operation alone would not identify mobility without it.

4. Sizing for $NM_H > NM_L$

For rail-to-rail static inverter outputs,

$$NM_H = V_{OH} - V_{IH} \approx V_{DD} - V_{IH}, \qquad NM_L = V_{IL} - V_{OL} \approx V_{IL}.$$

A stronger NMOS shifts the transfer curve left, decreasing both unity-slope input thresholds V_{IL}, V_{IH}. This increases NM_H and decreases NM_L. With equal threshold magnitudes the desired design is

$$\boxed{\beta_n > \beta_p, \qquad \beta_n = \mu_n C_{ox} \frac{W_n}{L_n}, \quad \beta_p = \mu_p C_{ox} \frac{W_p}{L_p}.}$$

Increase W_n/L_n relative to W_p/L_p after accounting for mobility; equal geometric sizes do not imply equal strengths. The midpoint switching voltage follows

$$V_M = \frac{V_{tn} + \sqrt{\beta_p/\beta_n}(V_{DD} - |V_{tp}|)}{1 + \sqrt{\beta_p/\beta_n}}.$$

The independent program also solves the two points where $dV_{out}/dV_{in} = -1$, using the correct saturation/linear current equations. For an illustrative ideal model ($V_{DD} = 1.8$ V, $V_{tn} = |V_{tp}| = 0.45$ V), it gives:

β_p/β_n	V_M (V)	V_{IL} (V)	NM_L (V)	NM_H (V)
0.5	0.822792	0.688420	0.688420	0.889328
1.0	0.900000	0.787500	0.787500	0.787500
2.0	0.977208	0.889328	0.889328	0.688420
4.0	1.050000	0.984360	0.984360	0.603557

At strength ratio 0.5, the strong NMOS gives $NM_H > NM_L$. At ratio 2, the strong PMOS gives the opposite relation. These are square-law calculations, not process-model measurements; threshold asymmetry in a real library should be included in its actual VTC.

Reconciliation and reproduction

The supplied handwritten `solution.jpg` was read before derivation. It agrees on the input-ramp explanation, the A–C critical path, and the mobility ratio under symmetric thresholds. Its noise-margin markings contain conflicting signs; the explicit definitions above settle the sign without relying on the correction marks. A right shift increases V_{IH} and therefore reduces NM_H.

Run `python verify.py`. It creates `clock_paths.csv`, `switching_points.csv`, and `verification.json`. The analytical calculations require only Python's standard library; the assignment does not require an electrical HSPICE run for these four questions.